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Original article

# Accuracy of bioimpedance equations for measuring body composition in a cohort of 2134 patients with obesity

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SUMMARY

*Introduction:* Measuring body composition is an important issue to phenotype patients with obesity and to follow the nutritional care efficiency. Bioelectrical Impedance Analysis (BIA) is a simple and rapid technique. However, validity of BIA in patients with obesity remains controversial. Thus, we aimed to evaluate the validity of several BIA equations to assess body composition in a large cohort of patients with obesity by using dual X ray absorptiometry (DXA) as reference.

*Methods:* Seven BIA equations have been retrospectively applied on electrical data measured by BIA in patients with obesity with BMI equal or higher than 30 kg/m$^2$ and results were compared to DXA-derived fat mass (FM) and fat-free mass (FFM). BIA and DXA were done the same day after an overnight fasting. Results were compared with Bland-Altman method and Pearson correlation. We also calculated the accuracy defined as the percentage of patients with DXA-BIA difference within ± 10% of DXA measures for FFM and FM.

*Results:* Data from 2134 patients with class I and II obesity (ob1/2, n = 1452, 47.4 ± 14.2 y; 35.0 ± 2.7 kg.m$^{-2}$) and class III obesity (ob3, n = 682, 48.2 ± 13.9 y; 44.1 ± 3.5 kg.m$^{-2}$) were analyzed. The best results to evaluate FFM both in ob1/2 and ob3 groups were obtained with Roubenoff's equation: Bland Altman bias at −1.96 and −0.82 kg, Pearson correlation r at 0.93 and 0.87, accuracy at 75.7% and 83.3%, respectively. However, limits of agreements at 95% were high: [-9.42; 5.49 kg] and [-8.16; 6.52 kg]. For FM evaluation, Roubenoff's equation also showed best results for ob1/2 group (bias at −1.17 kg; correlation r at 0.89 and accuracy at 79.1%) but not for ob3 group. In this latter group, Deurenberg's equation exhibited the best results (bias at 2.09 kg; correlation r at 0.81 and accuracy at 76.8%). However, limits of agreements remained high.

*Conclusion:* In patients with obesity, Roubenoff BIA equation should be recommended to assess fat free mass, even if limits of agreements remain high.

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## 1. Introduction

Body composition measure is used in clinical practice for nutritional status assessment and monitoring, particularly in patients with obesity to determine changes in fat mass (FM) and fat-free mass (FFM) after medical or surgical interventions, i.e. when body weight does not change but body composition does, or such as after bariatric surgery. Obesity treatment should not just reduce total body weight but achieve a reduction in FM while preserving FFM. In patients with obesity, low FFM combined with low skeletal

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muscle function, known as sarcopenic obesity [1], is associated with increased morbidity and mortality [2].

Several techniques have been developed for the assessment of body composition in human [3]. Dual-energy X-ray absorptiometry (DXA) is considered to be the reference method in clinical practice and research [4], and provides a rapid, non-invasive assessment of FM, FFM and bone mineral mass. But, DXA is expensive and requires specialised radiology equipment, thus hardly feasible in routine clinical practice. Bioelectrical Impedance Analysis (BIA) is commonly available and used in clinical practice and research studies. This method is rapid, simple, non-invasive, low-cost, and measures electrical data (resistance, reactance, impedance) of the body to a small alternating current [5]. Several BIA devices are available including systems with a single-frequency (SF-BIA) or Multiple frequencies-BIA (MF-BIA) current and systems requiring supine or stand-up position. From electrical data, predictive BIA equations involving the body's resistance, reactance or impedance to the flux, and other variables such as weight, height and age have been developed to calculate FFM or total body water [6]. However, BIA equations developed in a specific population are not generalizable to a population different from the validation sample, to avoid imprecise results and misinterpretation [7]. Many BIA equations have been developed in normal-weight subjects limiting the accuracy in patients with obesity. Moreover, body water distribution may be different in patients with obesity [8,9]. Comparison of body composition assessment by DXA and BIA according to the BMI has been poorly documented [10]. We previously compared body composition assessment by DXA and BIA according to the BMI in a large cohort of patients [11]. We found that for overweight and patients with obesity (25<BMI<40), BIA overestimated FFM from 7.18 to 8.28 kg, and underestimated FM from 4.02 to 5.67 kg compared with DXA. Surprisingly, DXA and BIA measures were very closed (difference < 1 kg) for FM in patients with BMI $\geq$ 40, while BIA overestimated FFM by 5.87 kg, and differences between DXA and BIA varied with BMI. However, in this latter study, we used BIA-derived body composition obtained by the manufacturer unknown algorithm. Very few studies have reported good concordance between the two methods [12,13] while many others have not [14–22], probably due to some limiting factors including the use of different BIA devices (SF-BIA, MF-BIA), small population size, differences in subject's age, ethnicity and body weights, but also different manufacturer equations. Indeed, in these latter studies, the unknown algorithms of manufacturers were used. To our knowledge, there was no study evaluating the validity of different BIA equations in patients with obesity.

Thus, in the present study, we aimed to evaluate the validity of several BIA equations to assess body composition in a large cohort of patients with obesity by using DXA as reference.

## 2. Material and methods

From 2010 to 2017, all adult patients with body mass index (BMI) higher than 30 kg.m$^{-2}$ and body composition assessment for routine clinical practice were retrospectively included. Patients with missing data (BIA, DXA, body weight, height, age) were excluded. After an overnight fasting period of 12 h, body weight and height were measured and used to calculate BMI (kg.m$^{-2}$). Dual-energy X-ray absorptiometry (DXA) and bioimpedance (BIA) were performed by the same operator. The study was approved by the Local Ethics Committee for Non-Interventional Studies (CERNI) and all data were fully anonymized.

### 2.1. Dual-energy X-ray absorptiometry (DXA)

DXA was performed on the whole body using a Lunar Prodigy Advance (General Electric Healthcare, enCORE software version 13.60) without specific preparation. The assessment of QA and QC data for the DXA measurements was done every morning when patient assessment was planned and at least 3 days per week. The manufacturer serviced the DXA equipment at least once per year. During measurement, all patients had their underwear, without metal accessories worn. DXA uses two X-ray beams with different energy levels. Based on their X-ray attenuation properties, FFM (lean mass and bone mineral content) and FM were measured.

### 2.2. Bioelectrical Impedance Analysis (BIA)

Body composition, FFM and FM, was assessed as previously described [23] in supine position using a MF-BIA, Bodystat Quadscan 4000 according to the manufacturer's recommendations. The accuracy and precision of the device was checked by using a manufacturer 's external circuit including a precision resistor and capacitor. The accepted coefficient of variation was less than 1% at each frequency. The BIA resistance and reactance were measured at 50 kHz, while BIA impedance was determined at each frequency (5, 50, 100 and 200 kHz).

Seven BIA equations (Table 1) were retrospectively applied to these electrical data measured by BIA to evaluate body composition [23–29]. The population studied to develop these equations, the method used as reference and the BIA devices are displayed in Table 1. Five equations calculated FFM. Another equation calculated total body water (BWa). In this latter case, FFM was then calculated taking into account FFM hydration at 73%. FM was then calculated by the difference between body weight and FFM.

### 2.3. Data analysis and statistics

Results obtained by DXA and BIA according to the equation used were compared by using Bland-Altman method leading to the bias, the confidence interval at 95% (95% CI) and the correlation coefficient of the bias. Pearson correlations were also performed and a p-value < 0.05 was considered significant. Then, the concordance coefficient of Lin [30] was calculated taking into account that a Lin's concordance coefficient or $\rho$ value $\geq$ 0.81 was considered as fairly good as previously described [31]. For FFM and FM, the accuracy of each BIA equation was calculated by the percentage of patients with a difference between BIA and DXA results within ± 10% or ± 5% of the data obtained by DXA.

## 3. Results

In the present study, we retrospectively included the data of 2134 patients with obesity followed in the Nutrition department of Rouen University Hospital between 2010 and 2017. These patients have been classified in two sub-groups: patients with class I and II obesity, n = 1452 (35.0 ± 2.7 kg.m$^{-2}$), and patients with class III obesity, n = 682 (44.1 ± 3.5 kg.m$^{-2}$). Demographic data and body composition measured by DXA are displayed in Table 2.

1. Comparison of DXA and BIA-derived values for fat mass and fat free mass in patients with class I and II obesity

Data of comparison are reported in Table 3 and Figs. 1–3. For FFM (49.6 ± 9.8 kg by DXA), Pearson correlation r was equal or greater than 0.8 for all BIA equations. The best r was observed for Kyle and Roubenoff equations. However, the concordance coefficient of Lin was greater than 0.81, which is considered as fairly good [30], only for Roubenoff's equation (0.90, Fig. 3). Other equations showed lower concordance coefficient of Lin, particularly Kyle equation with Lin $\rho$ at 0.39. Only the Roubenoff's equation showed a bias lower than ±10% (−1.96 kg, −5.1%). Other equations showed a

**Table 1**
Tested bioimpedance equations for body composition.

| | Population | Method used as reference | BIA device | Equation |
|---|---|---|---|---|
| Deurenberg et al. [24] | Healthy subjects | Densitometry | RJL-101 | FFM = −12.44 + (0.34 x height$^2$/Resistance) + (0.1534 x height) + (0.273 x weight) - (0.127 x age) + (4.229 x sex) |
| Kyle et al. [23] | Healthy subjects | DXA | Xitron | FFM = −4.104 + (0.518 x height$^2$/Resistance) + (0.231 x weight) + (0.130 x Reactance) + (4.229 x sex) |
| Sun et al. [25] | Healthy subjects | Densitometry | RJL-101 | FFM = [sex factor 1] + ([sex factor 2] x height$^2$/Resistance) + ([sex factor 3] x weight) + (0.016 x Resistance) |
| Roubenoff et al. [26] | Elderly people | DXA | RJL-101 | FFM = 5.741 + (0.4551 x height$^2$/Resistance) + (0.1405 x weight) + (0.0573 x Reactance) + (6.2467 x sex) |
| Lukaski et al. [28] | Healthy subjects | Densitometry | RJL-101 | FFM = −4.03 + (0.734 x height$^2$/Resistance + (0.116 x weight) + (0.096 x reactance) + (0.984 x sex) |
| Lohman et al. [29] | Healthy subjects | Densitometry | Valhalla | FFM = [factor 1] + ([factor 2] x height$^2$/Resistance) + ([factor 3] x weight) + ([factor 4] x reactance) |
| Kushner et al. [27] | Healthy subjects | $D_2O$ | RJL-101 | BWa = 8.315 + 0.382 x height$^2$/Resistance) + 0.105 x weight |

BWa, body water; FFM, fat-free mass. Sex: 1 for men, 0 for women. For Sun 's equation: sex factor 1 = −10.68 for men, −9.53 for women; sex factor 2 = 0.65 for men, 0.69 for women; sex factor 3 = 0.26 for men, 0.17 for women. For Lohman's equation: factors 1, 2, 3 and 4 are dependent of sex and age (See supplemental table S1). Population, methods used as reference and the BIA device for the development of the equation are detailed.

**Table 2**
Demographic data.

| | Class I and II obesity n = 1452 | Class III obesity n = 682 |
|---|---|---|
| Gender (F/M) | 1124/328 | 593/89 |
| Age (y) | 47.4 ± 14.2 | 45.2 ± 13.9 |
| Body weight (kg) | 96.2 ± 12.3 | 117.3 ± 11.4 |
| BMI (kg.m$^{-2}$) | 35.0 ± 2.7 | 44.1 ± 3.5 |
| Fat mass (kg)[a] | 43.6 ± 7.1 | 58.0 ± 7.4 |
| Fat mass (%) | 47.8 ± 6.5 | 52.0 ± 4.5 |
| Fat-free mass (kg)[a] | 49.6 ± 9.8 | 54.2 ± 7.7 |
| Fat-free mass (%) | 52.2 ± 6.7 | 48.0 ± 4.5 |

[a] Measured by DXA.

bias greater than ±10% from −12.4% to −21.2%. However, all tested BIA equations showed large 95% CI, as shown on Bland Altman plots (Fig. 1). In addition, we observed that the difference of FFM between BIA and DXA values increased for Sun (r = −0.35, p < 0.0001) and Lukaski (r = −0.16, p < 0.0001) equations with the increase of measured mean FFM while it decreased for Roubenoff (r = 0.19, p < 0.0001) and Kushner (r = 0.42, p < 0.0001) equations.

For FM (43.6 ± 7.1 kg), all equations showed a Pearson correlation r greater than 0.8, except Lohman equation. Kyle and Roubenoff equations showed the best r at 0.9. However, the Lin ρ was greater than 0.81 only for the Roubenoff's equation (0.88, Fig. 3) that also showed the lowest bias at −1.18 kg (−2.9%). Dereunberg and Lohman equations also showed a bias lower than ± 10% (−4.7% and 7.0%, respectively). However, 95% CI remained high for all equations as shown on the Bland Altman plots (Fig. 2). The difference of FM between BIA and DXA values markedly increased for Deurenberg's equation (r = 0.33, p < 0.0001) with the increase of measured mean FM and moderately for Kyle (r = 0.13, p < 0.0001) and Roubenoff (r = −0.14, p < 0.0001) equations.

By defining accuracy as the percentage of patients with a difference between BIA-derived and DXA values within ± 10% of DXA values, we observed that Roubenoff's equation provided the best results with 75.7 and 79.1% of patients for accurate FFM and accurate FM, respectively (Table 3 and Fig. 4A). Accurate assessment of other equations ranged from 10.7 to 43.8% of patients for FFM and from 23.6 to 66.2% of patients for FM. If we defined the accuracy within ± 5% of DXA values, the percentage of patients with accurate values with the Roubenoff's equation decreased from 75.7 to 45.2% for FFM and from 79.1 to 48.5% for FM (Table 3 and Fig. 4B) but Roubenoff's equation remained the equation providing the best results.

**Table 3**
Comparison of fat mass and fat-free mass values obtained by DXA and BIA according to the BIA equation used in patients with class I and II obesity.

| | Bias (kg) | 95% CI (kg) | Bias (%) | Accuracy (%) ±10% | Accuracy (%) ±5% | Pearson r | Lin ρ |
|---|---|---|---|---|---|---|---|
| ***Fat-free mass*** | | | | | | | |
| Deurenberg et al. | −5.52 | [-14.88; 3.80] | −12.4 | 43.8 | 23.0 | 0.89 | 0.76 |
| Kyle et al. | −8.01 | [-15.69; −0.33] | −17.6 | 20.3 | 7.0 | 0.93 | 0.39 |
| Sun et al. | −9.93 | [-18.81; −1.05] | −21.2 | 10.7 | 3.4 | 0.92 | 0.63 |
| Roubenoff et al. | −1.96 | [-9.42; 5.49] | −5.1 | 75.7 | 45.2 | 0.93 | 0.90 |
| Lukaski et al. | −7.72 | [-16.26; 0.82] | −16.8 | 26.0 | 9.6 | 0.92 | 0.70 |
| Lohman et al. | −6.41 | [-17.29; 4.45] | −14.6 | 41.3 | 19.8 | 0.85 | 0.69 |
| Kushner et al. | −7.71 | [-16.35; 0.93] | −17.6 | 24.8 | 12.0 | 0.91 | 0.64 |
| ***Fat mass*** | | | | | | | |
| Deurenberg et al. | 2.37 | [-5.30; 10.04] | 4.7 | 66.2 | 34.3 | 0.83 | 0.77 |
| Kyle et al. | 4.86 | [-1.20; 10.93] | 11.1 | 41.7 | 15.1 | 0.90 | 0.72 |
| Sun et al. | 6.78 | [-0.41; 13.99] | 15.7 | 24.4 | 7.8 | 0.87 | 0.60 |
| Roubenoff et al. | −1.18 | [-7.76; 5.41] | −2.9 | 79.1 | 48.5 | 0.90 | 0.88 |
| Lukaski et al. | 4.57 | [-2.65; 11.81] | 10.6 | 46.9 | 20.2 | 0.87 | 0.73 |
| Lohman et al. | 3.26 | [-6.58; 13.11] | 7.0 | 60.7 | 33.5 | 0.75 | 0.68 |
| Kushner et al. | 4.57 | [-3.29; 12.42] | 10.3 | 42.4 | 19.1 | 0.84 | 0.70 |

Bias was calculated by the difference [DXA value − BIA derived value]. Accuracy is the percentage of patients with DXA-BIA difference within ± 10% or ± 5% of DXA measure. 95% CI, 95% confidence interval.

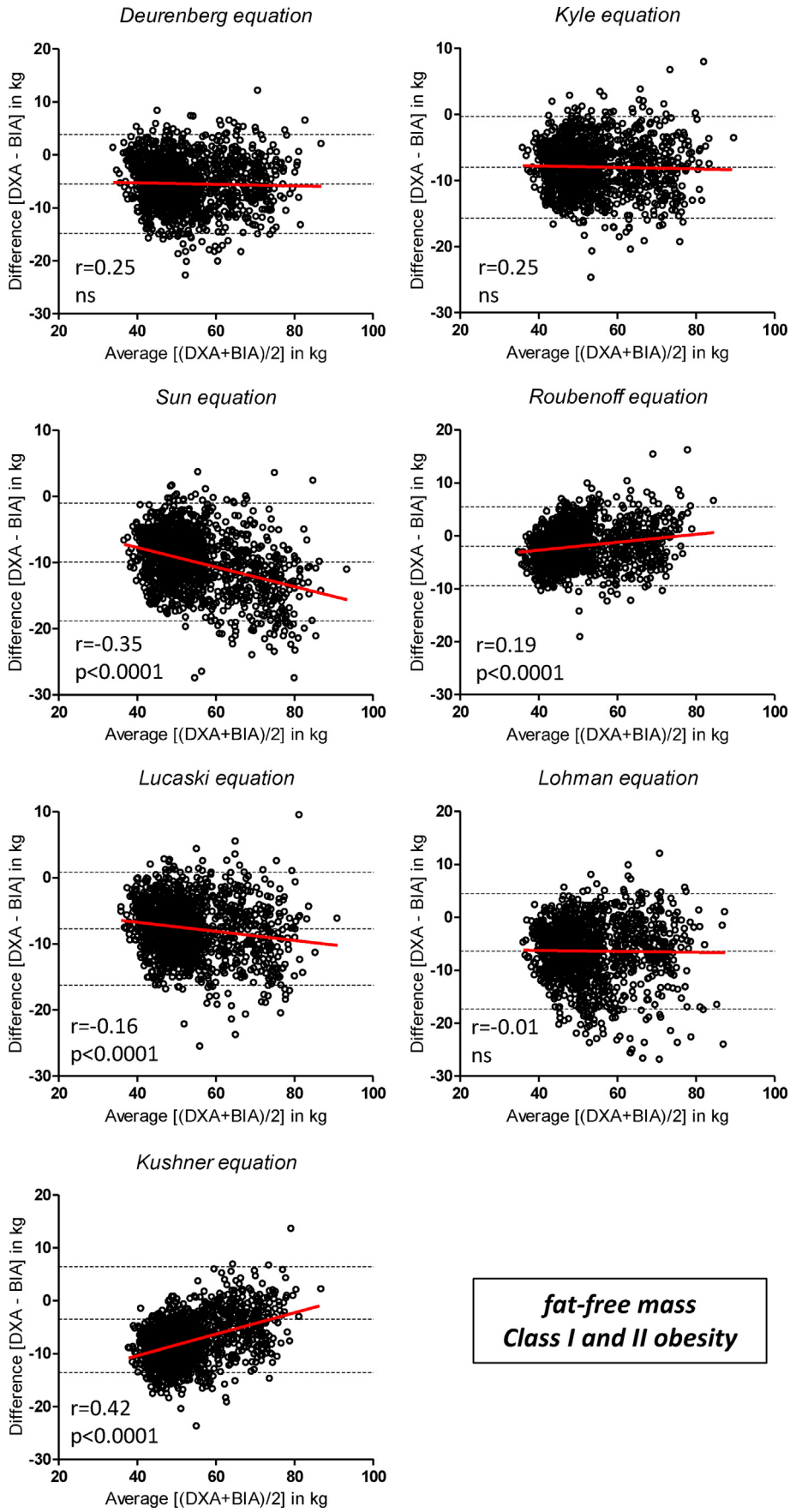


**Fig. 1.** Bland Altman plots for the comparison of fat-free mass (FFM) measured by DXA and BIA according to BIA equations. Values are obtained from 1452 patients with class I and II obesity. Correlation coefficients derived from Pearson correlation.

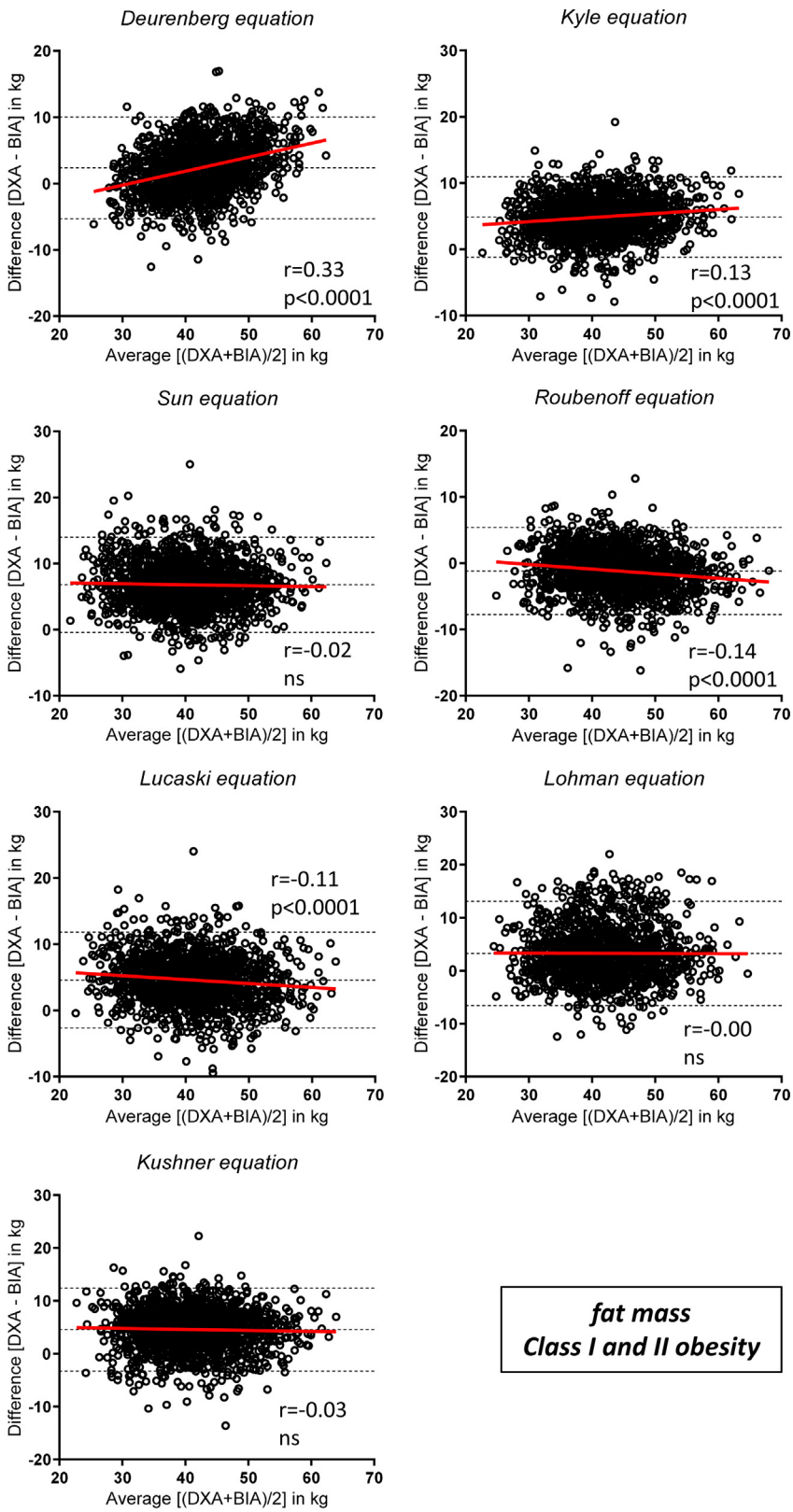


**Fig. 2.** Bland Altman plots for the comparison of fat mass (FM) measured by DXA and BIA according to BIA equations. Values are obtained from 1452 patients with class I and II obesity. Correlation coefficients derived from Pearson correlation.

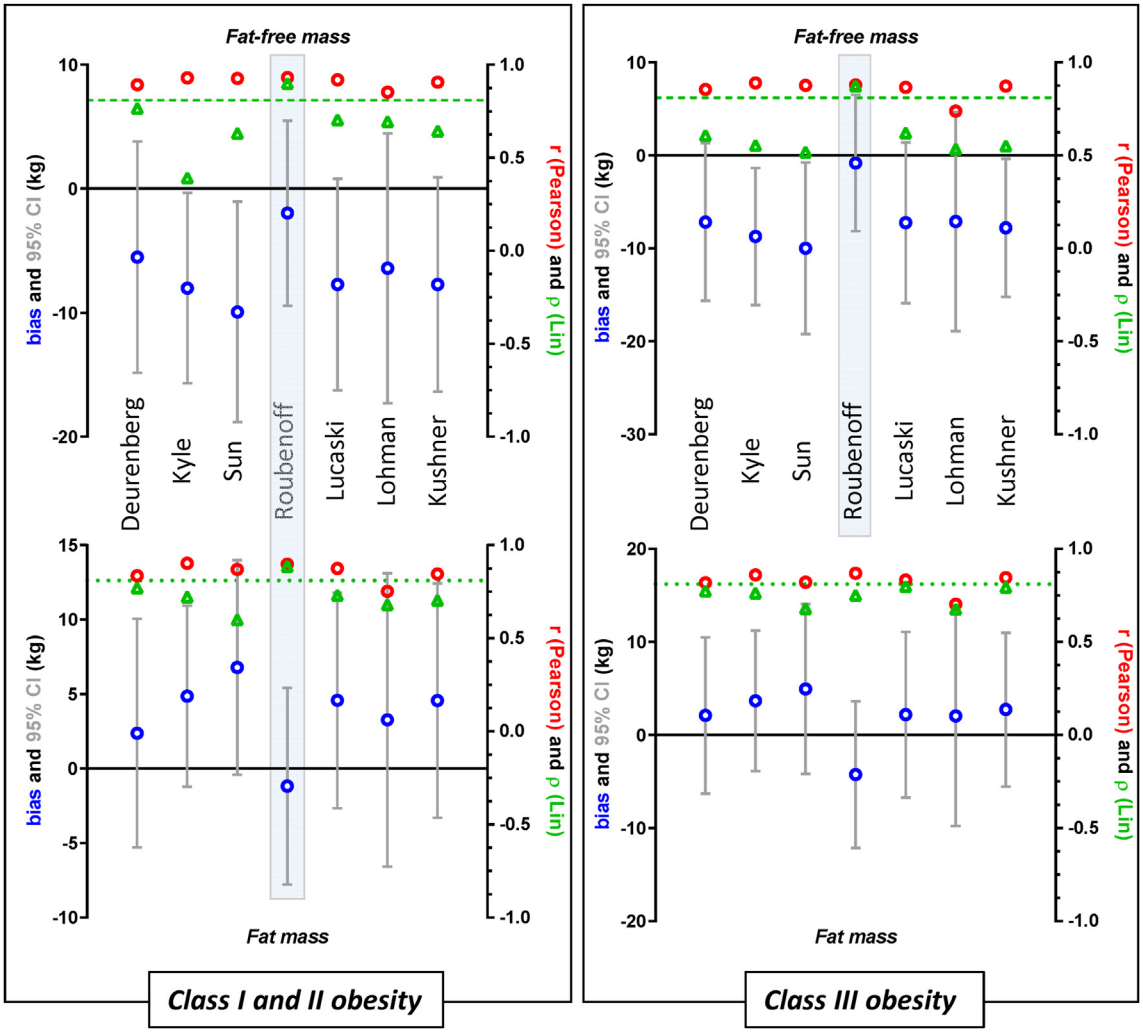


**Fig. 3.** Comparison of fat-free mass (FFM) and fat mass (FM) measured by DXA and BIA according to BIA equations. Fat-free mass (Upper panels) and fat mass (lower panels) values from 1452 patients with class I and II obesity (left panels) and 682 patients with class III obesity (right panels) were compared by Bland Altman test. Bias [DXA -BIA values] and 95% confidence intervals (95% CI) are shown in blue and grey, respectively. Pearson correlations were performed and Pearson r is shown in red circles. The Concordance coefficient of Lin ρ was calculated and is shown in green triangles. The green dashed line represents the cut-off at 0.81 to considered ρ as fairly good. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2. Comparison of DXA and BIA-derived values for fat mass and fat free mass in patients with class III obesity

Data of comparison are reported in Table 4, Figs. 3, 5 and 6. For FFM that has been evaluated by DXA at 54.2 ± 7.7 kg, all equations showed a Pearson correlation r greater than 0.8, except Lohman equation. The best r was observed for Kyle's equation (0.89)followed by Sun and Roubenoff equations at 0.88. Only the Roubenoff's equation showed a concordance coefficient of Lin greater than 0.81 at 0.87 (Fig. 3). Other equations showed low Lin ρ (from 0.52 to 0.62). Only the Roubenoff's equation showed a bias lower than ±10% at −0.82 kg (−1.92%). Other equations showed high bias from −13.7 to −18.7%. However, all tested BIA equations showed large 95% CI, as shown on the Bland Altman plots (Fig. 5). We also observed that the difference of FFM between DXA and BIA increased for Sun ($r = -0.42$, $p < 0.0001$), Lukaski ($r = -0.24$, $p < 0.0001$) and Lohman ($r = -0.17$, $p < 0.0001$) equations with the increase of measured mean FFM (Fig. 5) while it decreased for Kushner's equation ($r = 0.25$, $p < 0.0001$).

For FM (58.0 ± 7.4 kg by DXA), the Pearson correlation r was greater than 0.8 for all equations, except Lohman's equation. the best r was observed for Roubenoff's equation (0.9). However, for the evaluation of FM in patients with class III obesity, a Lin ρ greater than 0.81 was observed in any tested equations. Lukaski's equation showed the best Lin ρ at 0.80 (Fig. 3). All equations provided a bias lower than ± 10%. Dereunberg's equation showed a bias at 3.1% while the bias of Roubenoff's equation was −7.6%. Again, 95% CI remained high for all equations, as shown on the Bland Altman plots (Fig. 6). In addition, higher was the mean FM values, higher was the difference of FM values between DXA and BIA for Deurenberg ($r = 0.24$, $p < 0.0001$) and Roubenoff ($r = -0.17$, $p < 0.0001$) equations (Fig. 6).

Concerning the accuracy (Table 4 and Fig. 4A), we observed that Roubenoff's equation provided the best results with 83.3% of patients for FFM. Other equations showed very low accuracy from 13.6 to 43.5% for FFM. By contrast, for accurate FM, Roubenoff's equation did not provide the best results (66.7% of patients), which have been achieved by Dereunberg's equation with 76.8%. If we defined the accuracy within ± 5% of DXA values, the percentage of patients with accurate values with the Roubenoff's equation decreased from 83.3 to 53.1% for FFM and from 66.7 to 34.8% for FM (Table 4 and Fig. 4B). For FFM estimation, Roubenoff's equation

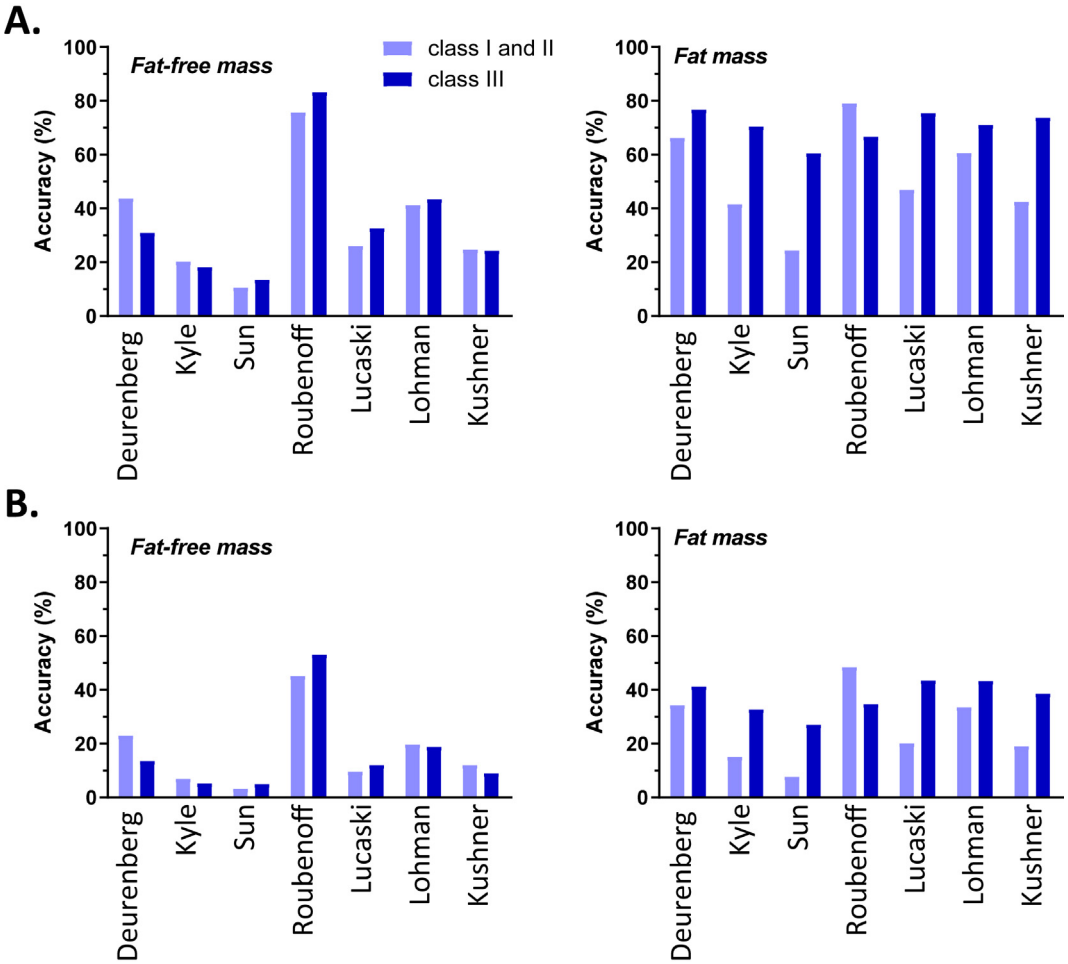


**Fig. 4.** Accuracy of BIA to measure fat-free mass (left panel) and fat mass (right panel) compared to DXA in 1452 patients with class I and II obesity and 682 patients with class III obesity. Accuracy was defined by the percentage of patients with a difference between BIA and DXA results within ± 10% (A.) or ± 5% (B.) of the data obtained by DXA.

**Table 4**
Comparison of fat mass and fat-free mass values obtained by DXA and BIA according to the BIA equation used in patients with class III obesity.

| | Bias | 95% CI | Bias | Accuracy (%) | | Pearson r | Lin ρ |
|---|---|---|---|---|---|---|---|
| | (kg) | (kg) | (%) | ±10% | ±5% | | |
| ***Fat-free mass*** | | | | | | | |
| Deurenberg et al. | −7.17 | [-15.63; 1.29] | −13.7 | 30.9 | 13.6 | 0.85 | 0.61 |
| Kyle et al. | −8.73 | [-16.10; −1.37] | −16.6 | 18.2 | 5.3 | 0.89 | 0.55 |
| Sun et al. | −10.01 | [-19.23; −0.77] | −18.7 | 13.6 | 5.0 | 0.88 | 0.52 |
| Roubenoff et al. | −0.82 | [-8.16; 6.52] | −1.92 | 83.3 | 53.1 | 0.88 | 0.87 |
| Lukaski et al. | −7.25 | [-15.89; 1.38] | −13.7 | 32.7 | 12.0 | 0.87 | 0.62 |
| Lohman et al. | −7.12 | [-18.92; 4.67] | −13.8 | 43.5 | 18.9 | 0.74 | 0.53 |
| Kushner et al. | −7.80 | [-15.24; −0.36] | −15.1 | 24.3 | 9.1 | 0.87 | 0.55 |
| ***Fat mass*** | | | | | | | |
| Deurenberg et al. | 2.09 | [-6.27; 10.47] | 3.1 | 76.8 | 41.3 | 0.82 | 0.77 |
| Kyle et al. | 3.66 | [-3.89; 11.21] | 6.1 | 70.5 | 32.8 | 0.86 | 0.76 |
| Sun et al. | 4.93 | [-4.19; 14.05] | 8.4 | 60.6 | 27.1 | 0.82 | 0.68 |
| Roubenoff et al. | −4.25 | [-12.13; 3.61] | −7.6 | 66.7 | 34.8 | 0.87 | 0.75 |
| Lukaski et al. | 2.18 | [-6.78; 11.06] | 3.6 | 75.5 | 43.5 | 0.83 | 0.80 |
| Lohman et al. | 2.04 | [-9.78; 13.87] | 3.2 | 71.1 | 43.4 | 0.70 | 0.67 |
| Kushner et al. | 2.72 | [-5.51; 10.96] | 4.5 | 73.7 | 38.6 | 0.84 | 0.79 |

Bias was calculated by the difference [DXA value − BIA derived value]. Accuracy is the percentage of patients with DXA-BIA difference within ± 10% or ± 5% of DXA measure. 95% CI, 95% confidence interval.

remained the equation providing the best results, the accuracy (±5%) for other equations ranging from 5.0 to 18.9%. For FM, Lukaski's equation provided the best percentage of accurate values (43.5%) while the percentage for Deurenberg's equation decreased to 41.3%.

3. Comparison of DXA and BIA-derived values for fat mass and fat free mass in patients with obesity according to the sex.

We also performed an analysis of our data according to the sex. Demographic data and body composition measured by DXA

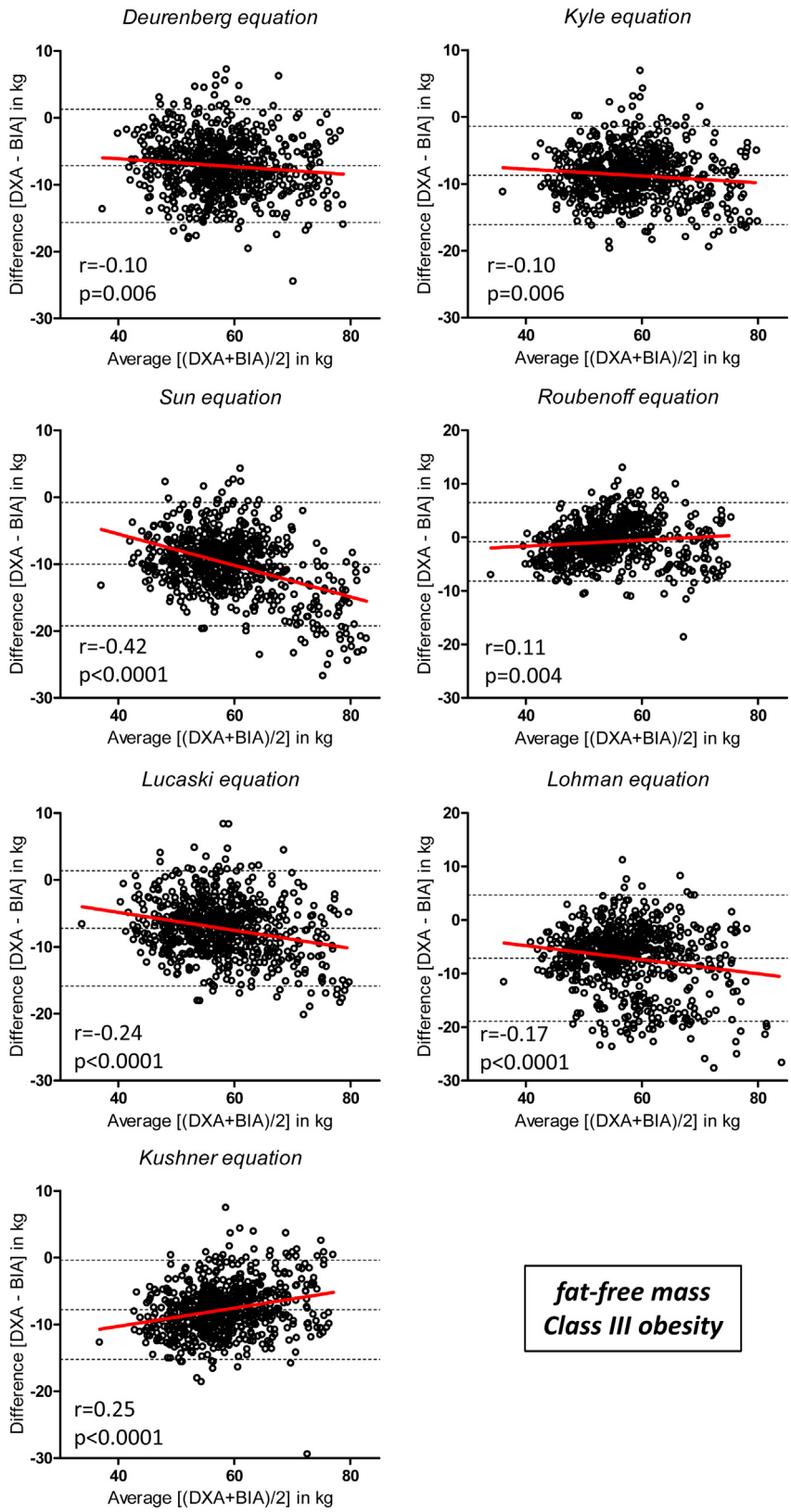


**Fig. 5.** Bland Altman plots for the comparison of fat-free mass (FFM) measured by DXA and BIA according to BIA equations. Values are obtained from 682 patients with class III obesity. Correlation coefficients derived from Pearson correlation.

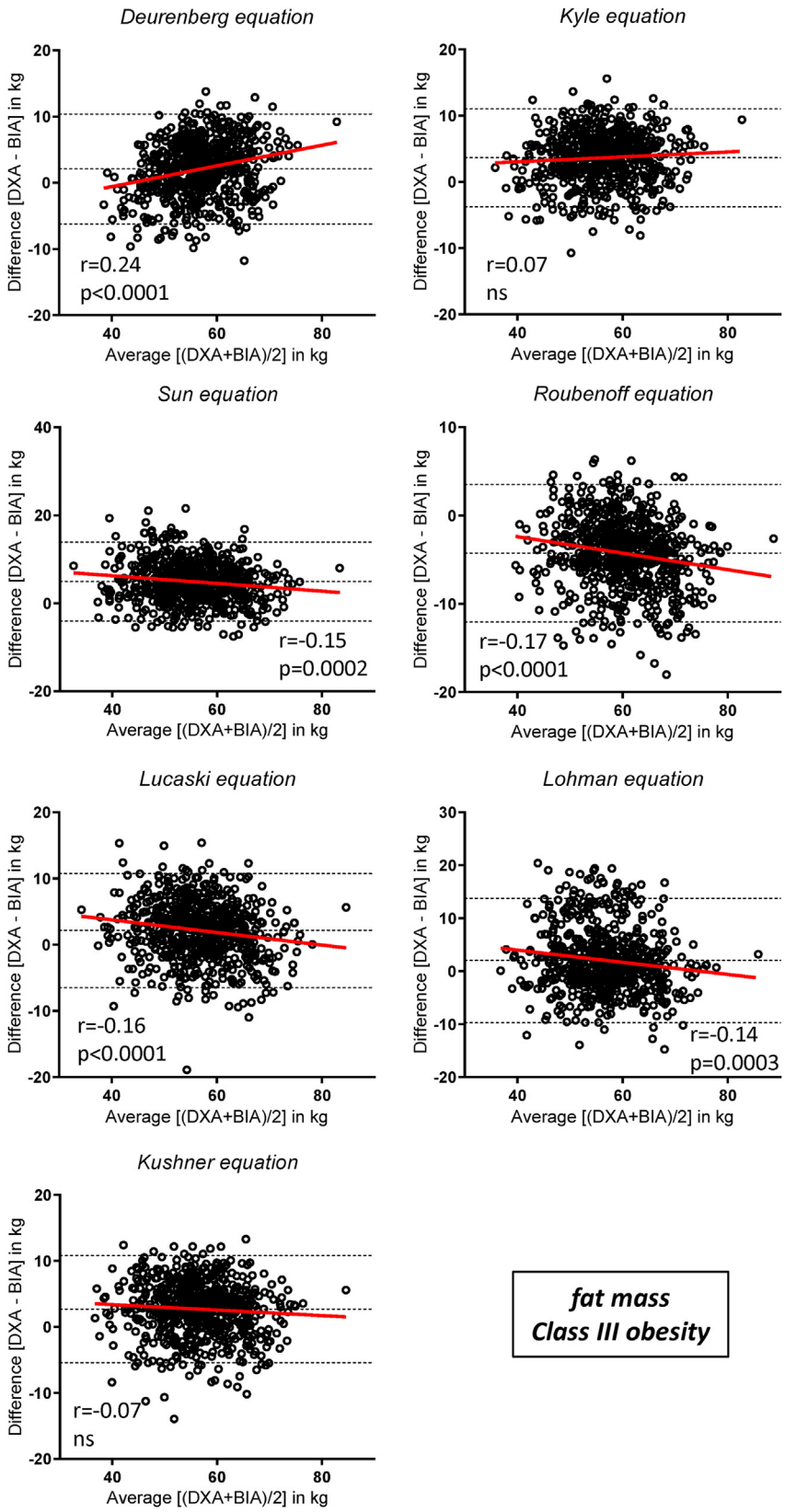


**Fig. 6.** Bland Altman plots for the comparison of fat mass (FM) measured by DXA and BIA according to BIA equations. Values are obtained from 682 patients with class III obesity. Correlation coefficients derived from Pearson correlation.

according to the sex are displayed in Supplemental Table S2. In our retrospective study, female patients are predominant. Male patients exhibited higher body weight than females but lower fat mass percentage measured by DXA. The Bland Altman plots for female and male patients with ob1/2 or ob3 are shown in Supplemental Figures S1-S8 and the data of comparison and accuracy are displayed in the Supplemental Tables S3-S6. Briefly, both in female and males patients, Roubenoff's equation provided the best results for FFM in patients with class I and II obesity but also in patients with class III obesity. Indeed, the percentage of accurate BIA measures (±10%) is 73.7 and 83.8% in female patients with class I and II obesity and class III obesity, respectively. Similarly, it was 82.6 and 80.9% for male patients, respectively. Other equations gave lower values of accuracy ranging for instance between 39.2 and 31.0% for Deurenberg's equation in female patients. For FM, Roubenoff's equation remained the best equation only for female patients with class I and II obesity (82.6% of accuracy), while Lukaski's equation exhibited the best accuracy in patients with class III obesity. In male patients, both in ob1/2 and ob3 groups, the best results for FM were achieved by the Kushner's equation but the sample size in male population was limited, particularly for ob3 group.

## 4. Discussion

In the present study, we reported for the first time the validity of several BIA equations to assess body composition in a large cohort of patients with obesity (n = 2134) by using DXA as reference. Our results suggest that Roubenoff's equation should be recommended to assess FFM in patients with obesity, whatever the severity of obesity, at least for studies at the population level. To estimate FM, Roubenoff's equation also seems the most appropriate equation at the population level but only in patients with class I and II obesity.

BIA is a rapid and non-invasive method to estimate body composition but its validity in extreme BMI remains debated. We previously reported in 115 female patients with anorexia nervosa (AN, 14.6 ± 1.2 kg/m$^2$) that none of the BIA tested equations was adapted to evaluate FFM and ALM, at the individual level [32] in accordance with another study [33]. Even if previous studies failed to show good agreements between BIA-based body composition and DXA values in subjects with obesity [18–22,34], none of these studies assessed the impact of the used BIA equations. Indeed, these studies used different BIA devices and the algorithms of the manufacturers that remains unknown for property reasons. To our knowledge, our study reports for the first time the validity of different BIA equations in subjects with obesity. In addition, another strength of our study is the large cohort of subjects (>2000 subjects) compared to previous studies.

Regarding FFM, the best results were obtained with Roubenoff's equation both in ob1/2 and ob3 groups reporting low bias (−1.96 and −0.82 kg respectively), acceptable accuracy at 10% (75.7% and 83.3%), good correlation (Pearson r at 0.93 and 0.87 respectively, Supplemental Table S7). However, limits of agreements were high [−9.42; 5.49 kg] and [−8.16; 6.52 kg] respectively. Regarding FM, Roubenoff's equation also showed best results for ob1/2 group (bias at −1.17 kg; correlation r at 0.89 and accuracy at 79.1%) but not for ob3 group. In this latter group, Deurenberg's equation exhibited the best results (bias at 2.09 kg; correlation r at 0.81 and accuracy at 76.8%). However, again, the limits of agreements were high. The wide limits of agreements whatever the tested equations highlight the limitation of the use of BIA-based body composition values at the individual level. However, Roubenoff's equation seems to be the most adapted equation for study at the population level. In addition, we observed an increase of the difference between BIA-based and DXA values for FM with the increase of obesity severity, whatever the tested equations. These data are in accordance with a previous study in adolescents with obesity [35]. The lack of a BIA obesity-specific equation validated for the estimation of body composition at the individual level underlines the need for further investigations to develop a new equation or algorithm. Finally, in our study, we enrolled both female and male patients. We are, thus, able to show that these results were not dependent of sex.

Both FFM and FM compartments are altered during obesity. The use of body composition measurement allows the monitoring of compartmental weight loss during interventions. The accuracy of BIA to measure body composition changes remains debated [13,36,37]. Further studies should thus assess the validity of BIA equations to monitor body composition changes. More recently, some studies tried to evaluate the accuracy of BIA to measure segmental body composition in patients with obesity such as visceral fat or appendicular lean mass (ALM) that are relevant for clinical practice to assess, respectively, the cardiometabolic risk and the complications associated to sarcopenia. Compared to DXA [38] or to CT scanner [39], the authors reported a lack of sensitivity of BIA for the estimation of visceral fat mass. Similarly, the estimation of ALM in 115 patients with obesity [40], using, respectively, the original proprietary equation of the device and the equations proposed by Kyle [41], Sergi [42], and Yamada [43] showed good correlation ($r = 0.925$, $p < 0.001$), low bias (−5.2%), but high limits of agreements [−14.0, +7.3 kg] compared to DXA. A recent review reported that none of the 25 existing BIA-equations to estimate muscle mass had been developed in subjects with obesity [44]. Further studies should thus evaluate both the validity of BIA equations from following body composition changes after medical or surgical interventions in patients with obesity and the predictive values of BIA-based body composition for metabolic disorders (fat mass), physical fitness (fat free mass) or other comorbidities.

The main strength of our study is the sample size (n = 2134) with 1452 patients with class I and II obesity of which 682 patients with class III obesity, who can be candidates to bariatric surgery. In addition, our study enrolled both female and male subjects and is the first one to assess the validity of different BIA equations in subjects with obesity. Our study has also some limitations: In the present work, even if BIA device calibration was checked by measuring impedance at each frequency, resistance and reactance are not directly controlled. In addition, we have tested BIA equations developed in non-obese healthy population by using different methods as reference, i.e. $D_2O$, DXA or densitometry. Interestingly, Roubenoff's equation that has been developed in elderly subjects provided the best results. We also evaluated the accuracy of two other equations developed in anorectic patients, the Scalfi [45] and *Bedogni* [46] equations, for which data are displayed in supplemental table S8. Another limitation is related to the estimation of fat mass by BIA (difference between body weight and BIA-based fat free mass value) that lead to an error amplification and thus to increase limits of agreements. Finally, in the present study, we did not consider hydration status of the studied population while it is well established that hydration status affect BIA results [8]. Studies have shown that patients with obesity exhibit variation of soft tissue hydration [47,48]. Indeed, Stoklossa et al. reviewed twelve studies investigating BIA in patients with obesity and concluded that an increase of total body water or extracellular water was associated with overestimation of fat-free mass [49]. Brunani et al. recently reported that the BIA vector ellipses was different in patients with obesity compared to the reference population [50]. In another study, 88.4% of patients with obesity exhibit altered hydration status [51], as defined in [52]. These data underlie that the use of vector length and phase angle may be of interest to estimate hydration status in patients with obesity as described in other conditions [53]. However, in the present

study, we have thus chosen to evaluate the validity of BIA equations in all patients with obesity whatever their hydration status.

## 5. Conclusion

In conclusions, the comparison of DXA and BIA suggests that DXA should remain the method of reference for measuring body composition in extreme BMI. BIA and DXA are not interchangeable techniques at the individual level, but BIA can be considered an easy, non-invasive, and low-cost method for body composition assessment in people with obesity, if we consider the error associated with the measurements, and select an adequate equation (Supplemental Table S7). The Roubenoff's equation should be recommended for studies at the population level. Further studies are needed to develop new BIA specific equations to evaluate body composition at the individual level in subjects with obesity. During patient's follow-up, when the same device and measurement protocol is used, BIA may be applicable for repeated measures to observe changes in body composition but this need also to be further evaluated.

## Author's contribution

MC, NA: study concept and design; MC, CL, VF, HL, SG, PD, NA: acquisition of data; MC, ME, CL, NA: analysis and interpretation of data; MC, NA: drafting of the manuscript; all authors: critical revision of the manuscript for important intellectual content; MC, ME: statistical analysis; MC, NA: study supervision.

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## Conflicts of interest

MC is shareholder of targEDys company. PD is a co-founder and shareholder of TargEDys company.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.clnu.2022.07.032.

## References

[1] Donini LM, Busetto L, Bauer JM, Bischoff S, Boirie Y, Cederholm T, et al. Critical appraisal of definitions and diagnostic criteria for sarcopenic obesity based on a systematic review. Clin Nutr 2020;39(8):2368–88.
[2] Prado CM, Wells JC, Smith SR, Stephan BC, Siervo M. Sarcopenic obesity: a Critical appraisal of the current evidence. Clin Nutr 2012;31(5):583–601.
[3] Ceniccola GD, Castro MG, Piovacari SMF, Horie LM, Correa FG, Barrere APN, et al. Current technologies in body composition assessment: advantages and disadvantages. Nutrition 2019;62:25–31.
[4] Shepherd JA, Ng BK, Sommer MJ, Heymsfield SB. Body composition by DXA. Bone 2017;104:101–5.
[5] Jackson AA, Johnson M, Durkin K, Wootton S. Body composition assessment in nutrition research: value of BIA technology. Eur J Clin Nutr 2013;67(Suppl 1): S71–8.
[6] Kyle UG, Bosaeus I, De Lorenzo AD, Deurenberg P, Elia M, Gomez JM, et al. Bioelectrical impedance analysis–part I: review of principles and methods. Clin Nutr 2004;23(5):1226–43.
[7] Blue MNM, Tinsley GM, Ryan ED, Smith-Ryan AE. Validity of body-composition methods across racial and ethnic populations. Adv Nutr 2021;12(5):1854–62.
[8] Deurenberg P. Limitations of the bioelectrical impedance method for the assessment of body fat in severe obesity. Am J Clin Nutr 1996;64(3 Suppl). 449S-52S.
[9] Waki M, Kral JG, Mazariegos M, Wang J, Pierson Jr RN, Heymsfield SB. Relative expansion of extracellular fluid in obese vs. nonobese women. Am J Physiol 1991;261(2 Pt 1):E199–203.
[10] Schiavo L, Pilone V, Tramontano S, Rossetti G, Iannelli A. May bioelectrical impedance analysis method Be used in alternative to the dual-energy X-ray absorptiometry in the assessment of fat mass and fat-free mass in patients with obesity? Pros, cons, and perspectives. Obes Surg 2020;30(8):3212–5.
[11] Achamrah N, Colange G, Delay J, Rimbert A, Folope V, Petit A, et al. Comparison of body composition assessment by DXA and BIA according to the body mass index: a retrospective study on 3655 measures. PLoS One 2018;13(7): e0200465.
[12] Stewart SP, Bramley PN, Heighton R, Green JH, Horsman A, Losowsky MS, et al. Estimation of body composition from bioelectrical impedance of body segments: comparison with dual-energy X-ray absorptiometry. Br J Nutr 1993;69(3):645–55.
[13] Thomson R, Brinkworth GD, Buckley JD, Noakes M, Clifton PM. Good agreement between bioelectrical impedance and dual-energy X-ray absorptiometry for estimating changes in body composition during weight loss in overweight young women. Clin Nutr 2007;26(6):771–7.
[14] Panotopoulos G, Ruiz JC, Guy-Grand B, Basdevant A. Dual x-ray absorptiometry, bioelectrical impedance, and near infrared interactance in obese women. Med Sci Sports Exerc 2001;33(4):665–70.
[15] Leahy S, O'Neill C, Sohun R, Jakeman P. A comparison of dual energy X-ray absorptiometry and bioelectrical impedance analysis to measure total and segmental body composition in healthy young adults. Eur J Appl Physiol 2012;112(2):589–95.
[16] Mattar L, Godart N, Melchior JC, Falissard B, Kolta S, Ringuenet D, et al. Underweight patients with anorexia nervosa: comparison of bioelectrical impedance analysis using five equations to dual X-ray absorptiometry. Clin Nutr 2011;30(6):746–52.
[17] Gaba A, Kapus O, Cuberek R, Botek M. Comparison of multi- and single-frequency bioelectrical impedance analysis with dual-energy X-ray absorptiometry for assessment of body composition in post-menopausal women: effects of body mass index and accelerometer-determined physical activity. J Hum Nutr Diet – Off J Br Diet Assoc 2015;28(4):390–400.
[18] Nickerson BS, McLester CN, McLester JR, Kliszczewicz BM. Agreement between 2 segmental bioimpedance devices, BOD POD, and DXA in obese adults. J Clin Densitom 2020;23(1):138–48.
[19] Wang ZH, Yang ZP, Wang XJ, Dong YH, Ma J. Comparative analysis of the multi-frequency bio-impedance and dual-energy X-ray absorptiometry on body composition in obese subjects. Biomed Environ Sci 2018;31(1):72–5.
[20] Ellegard L, Bertz F, Winkvist A, Bosaeus I, Brekke HK. Body composition in overweight and obese women postpartum: bioimpedance methods validated by dual energy X-ray absorptiometry and doubly labeled water. Eur J Clin Nutr 2016;70(10):1181–8.
[21] Neovius M, Hemmingsson E, Freyschuss B, Udden J. Bioelectrical impedance underestimates total and truncal fatness in abdominally obese women. Obesity 2006;14(10):1731–8.
[22] Pateyjohns IR, Brinkworth GD, Buckley JD, Noakes M, Clifton PM. Comparison of three bioelectrical impedance methods with DXA in overweight and obese men. Obesity 2006;14(11):2064–70.
[23] Kyle UG, Genton L, Karsegard L, Slosman DO, Pichard C. Single prediction equation for bioelectrical impedance analysis in adults aged 20–94 years. Nutrition 2001;17(3):248–53.
[24] Deurenberg P, van der Kooy K, Leenen R, Weststrate JA, Seidell JC. Sex and age specific prediction formulas for estimating body composition from bioelectrical impedance: a cross-validation study. Int J Obes 1991;15(1):17–25.
[25] Sun SS, Chumlea WC, Heymsfield SB, Lukaski HC, Schoeller D, Friedl K, et al. Development of bioelectrical impedance analysis prediction equations for body composition with the use of a multicomponent model for use in epidemiologic surveys. Am J Clin Nutr 2003;77(2):331–40.
[26] Roubenoff R, Baumgartner RN, Harris TB, Dallal GE, Hannan MT, Economos CD, et al. Application of bioelectrical impedance analysis to elderly populations. J Gerontol Biol Med Sci 1997;52(3):M129–36.
[27] Kushner RF, Schoeller DA. Estimation of total body water by bioelectrical impedance analysis. Am J Clin Nutr 1986;44(3):417–24.
[28] Lukaski HC, Bolonchuk WW, Hall CB, Siders WA. Validation of tetrapolar bioelectrical impedance method to assess human body composition. J Appl Physiol 1986;60(4):1327–32 (1985).
[29] Lohman T. Advances in body composition assessment. Champaign, IL: Human kinetics Publishers; 1992.
[30] Lin LI. A concordance correlation coefficient to evaluate reproducibility. Biometrics 1989;45(1):255–68.
[31] Partik BL, Stadler A, Schamp S, Koller A, Voracek M, Heinz G, et al. 3D versus 2D ultrasound: accuracy of volume measurement in human cadaver kidneys. Invest Radiol 2002;37(9):489–95.
[32] Coeffier M, Gate M, Rimbert A, Petit A, Folope V, Grigioni S, et al. Validity of bioimpedance equations to evaluate fat-free mass and muscle mass in severely malnourished anorectic patients. J Clin Med 2020;9(11).

[33] Marra M, Sammarco R, De Filippo E, Caldara A, Speranza E, Scalfi L, et al. Prediction of body composition in anorexia nervosa: results from a retrospective study. Clin Nutr 2018;37(5):1670–4.
[34] Cruz Rivera PN, Goldstein RL, Polak M, Lazzari AA, Moy ML, Wan ES. Performance of bioelectrical impedance analysis compared to dual X-ray absorptiometry (DXA) in Veterans with COPD. Sci Rep 2022;12(1):1946.
[35] Verney J, Metz L, Chaplais E, Cardenoux C, Pereira B, Thivel D. Bioelectrical impedance is an accurate method to assess body composition in obese but not severely obese adolescents. Nutr Res 2016;36(7):663–70.
[36] Thivel D, Verney J, Miguet M, Masurier J, Cardenoux C, Lambert C, et al. The accuracy of bioelectrical impedance to track body composition changes depends on the degree of obesity in adolescents with obesity. Nutr Res 2018;54:60–8.
[37] Verdich C, Barbe P, Petersen M, Grau K, Ward L, Macdonald I, et al. Changes in body composition during weight loss in obese subjects in the NUGENOB study: comparison of bioelectrical impedance vs. dual-energy X-ray absorptiometry. Diabetes Metab 2011;37(3):222–9.
[38] Kumar K, Sharma R, Manrai M, Sood AK. Visceral adipose tissue as a risk factor for diabetes mellitus in patients with chronic pancreatitis: a cross-sectional, observational study. Diabetes Ther 2017;8(5):1057–64.
[39] Xu Z, Liu Y, Yan C, Yang R, Xu L, Guo Z, et al. Measurement of visceral fat and abdominal obesity by single-frequency bioelectrical impedance and CT: a cross-sectional study. BMJ Open 2021;11(10):e048221.
[40] Ballesteros-Pomar MD, Gonzalez-Arnaiz E, Pintor-de-la Maza B, Barajas-Galindo D, Ariadel-Cobo D, Gonzalez-Roza L, et al. Bioelectrical impedance analysis as an alternative to dual-energy x-ray absorptiometry in the assessment of fat mass and appendicular lean mass in patients with obesity. Nutrition 2021;93:111442.
[41] Kyle UG, Genton L, Hans D, Pichard C. Validation of a bioelectrical impedance analysis equation to predict appendicular skeletal muscle mass (ASMM). Clin Nutr 2003;22(6):537–43.
[42] Sergi G, De Rui M, Veronese N, Bolzetta F, Berton L, Carraro S, et al. Assessing appendicular skeletal muscle mass with bioelectrical impedance analysis in free-living Caucasian older adults. Clin Nutr 2015;34(4):667–73.
[43] Yamada Y, Nishizawa M, Uchiyama T, Kasahara Y, Shindo M, Miyachi M, et al. Developing and validating an age-independent equation using multi-frequency bioelectrical impedance analysis for estimation of appendicular skeletal muscle mass and establishing a cutoff for sarcopenia. Int J Environ Res Publ Health 2017;14(7).
[44] Beaudart C, Bruyere O, Geerinck A, Hajaoui M, Scafoglieri A, Perkisas S, et al. Equation models developed with bioelectric impedance analysis tools to assess muscle mass: a systematic review. Clin Nutr ESPEN 2020;35:47–62.
[45] Scalfi L, Bedogni G, Marra M, Di Biase G, Caldara A, Severi S, et al. The prediction of total body water from bioelectrical impedance in patients with anorexia nervosa. Br J Nutr 1997;78(3):357–65.
[46] Bedogni G, Marra M, Bianchi L, Malavolti M, Nicolai E, De Filippo E, et al. Comparison of bioelectrical impedance analysis and dual-energy X-ray absorptiometry for the assessment of appendicular body composition in anorexic women. Eur J Clin Nutr 2003;57(9):1068–72.
[47] Das SK, Roberts SB, Kehayias JJ, Wang J, Hsu LK, Shikora SA, et al. Body composition assessment in extreme obesity and after massive weight loss induced by gastric bypass surgery. Am J Physiol Endocrinol Metab 2003;284(6):E1080–8.
[48] Resende CM, Camelo Junior JS, Vieira MN, Ferriolli E, Pfrimer K, Perdona GS, et al. Body composition measures of obese adolescents by the deuterium oxide dilution method and by bioelectrical impedance. Braz J Med Biol Res 2011;44(11):1164–70.
[49] Johnson Stoklossa CA, Forhan M, Padwal RS, Gonzalez MC, Prado CM. Practical considerations for body composition assessment of adults with class II/III obesity using bioelectrical impedance analysis or dual-energy X-ray absorptiometry. Curr Obes Rep 2016;5(4):389–96.
[50] Brunani A, Perna S, Soranna D, Rondanelli M, Zambon A, Bertoli S, et al. Body composition assessment using bioelectrical impedance analysis (BIA) in a wide cohort of patients affected with mild to severe obesity. Clin Nutr 2021;40(6):3973–81.
[51] Nicoletti CF, Camelo Jr JS, dos Santos JE, Marchini JS, Salgado Jr W, Nonino CB. Bioelectrical impedance vector analysis in obese women before and after bariatric surgery: changes in body composition. Nutrition 2014;30(5):569–74.
[52] Piccoli A, Nigrelli S, Caberlotto A, Bottazzo S, Rossi B, Pillon L, et al. Bivariate normal values of the bioelectrical impedance vector in adult and elderly populations. Am J Clin Nutr 1995;61(2):269–70.
[53] Massari F, Iacoviello M, Scicchitano P, Mastropasqua F, Guida P, Riccioni G, et al. Accuracy of bioimpedance vector analysis and brain natriuretic peptide in detection of peripheral edema in acute and chronic heart failure. Heart Lung 2016;45(4):319–26.